

CSE 211 (Theory of Computation)

Atif Hasan Rahman

Assistant Professor
Department of Computer Science and Engineering
Bangladesh University of Engineering & Technology

Adapted from slides by
Dr. Muhammad Masroor Ali

Theory of Computation

- Three areas of theory of computation
 - Automata
 - Computability
 - Complexity
- Linked by the question
 - What are the fundamental capabilities and limitations of computers?

Theory of Computation

- Automata
 - Automaton - a machine made in imitation of a human being
 - DFA, NFA
 - Context-free grammar (CFG), pushdown automata (PDA)
- Computability
 - Decidability
 - What can or cannot be solved
- Complexity
 - Tractability
 - What can or cannot be solved “efficiently”
 - Time complexity: P, NP, NP-complete, NP-hard
 - Space complexity: PSPACE

Theory of Computation

“Computer science is no more about computers than astronomy is about telescopes. ”

— Edsger W. Dijkstra

Theory of Computation

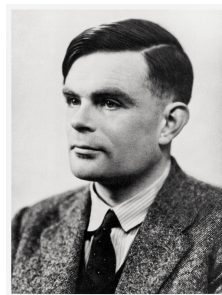
- Computation and computability
 - Building fast vehicles vs can we exceed the speed of light?
 - Building efficient engines vs can we build a perpetual motion machine?

Models of Computation

- Finite Automata
 - DFA, NFA
 - Limited amount of memory
 - Applications in compilers, control units of hardware, etc.
- Context-free grammar
 - More expressive than finite automata
 - Applications in compilers, AI and many other areas
- Turing Machine
 - Even more powerful
 - Can simulate a computer!
 - Problems Turing machine cannot solve are beyond theoretical limits of computation

Alan Turing

- “father of theoretical computer science and artificial intelligence” - wiki
- Proposed Turing machines
 - A general model of computation
 - Can simulate a computer
- Helped break the Enigma code during WW II
- Proposed the Turing test for AI
 - Distinguishing humans and computers through interrogation



Syllabus

- Regular languages
 - Regular expressions
- Finite automata
 - Deterministic finite automata (DFA)
 - Nondeterministic finite automata (NFA)
- Context-free languages
 - Context free grammars (CFG)
 - Pushdown automata (PDA)
- Turing machines
 - Equivalence with a computer
- Decidability
- Complexity
 - Time and space complexity

Syllabus

- Regular languages
 - Regular expressions
- Finite automata
 - Deterministic finite automata (DFA)
 - Nondeterministic finite automata (NFA)
- Context-free languages
 - Context free grammars (CFG)
 - Pushdown automata (PDA)
- Turing machines
 - Equivalence with a computer
- Decidability
- Complexity
 - Time and space complexity

Syllabus

- Regular languages
 - Regular expressions
- Finite automata
 - Deterministic finite automata (DFA)
 - Nondeterministic finite automata (NFA)
- Context-free languages
 - Context free grammars (CFG)
 - Pushdown automata (PDA)
- Turing machines
 - Equivalence with a computer
- Decidability
- Complexity
 - Time and space complexity

Regular Languages

- Regular languages
 - Languages recognized by finite automata - DFA, NFA
 - Languages described by regular expressions
- Limitations
 - Finite number of states
 - Hence finite amount of memory
- An example of a non-regular language

Regular Languages

- Regular languages
 - Languages recognized by finite automata - DFA, NFA
 - Languages described by regular expressions
- Limitations
 - Finite number of states
 - Hence finite amount of memory
- An example of a non-regular language
 - $B = \{0^n 1^n | n \geq 0\}$

Regular Languages

- Regular languages
 - Languages recognized by finite automata - DFA, NFA
 - Languages described by regular expressions
- Limitations
 - Finite number of states
 - Hence finite amount of memory
- An example of a non-regular language
 - $B = \{0^n 1^n | n \geq 0\}$

Context-Free Languages

- Context-Free Languages
 - Languages described by context-free grammars (CFG)
 - Languages recognized by pushdown automata (PDA)
- Extensively used in compilers (parsers)
- First used in study of human languages

An Informal Example

- Language of palindromes
- Palindrome
 - A string that reads the same backward and forward
 - 0110, 11011, ϵ
- Recursive definition for palindromes (over binary alphabet)
 - 0, 1 and ϵ are palindromes
 - if w is a palindrome, then $0w0$ and $1w1$ are palindromes

An Informal Example

aibohphobia

- The word used for the *fear of palindromes*. The word *aibohphobia* is itself a palindrome
- `https://www.urbandictionary.com/define.php?term=aibohphobia`

An Informal Example

- A CFG for palindromes

- $P \rightarrow \epsilon$
- $P \rightarrow 0$
- $P \rightarrow 1$
- $P \rightarrow 0P0$
- $P \rightarrow 1P1$

Exercise

- Design a CFG for the language

$$\{0^n 1^n \mid n \geq 0\}$$

- ϵ
- 01
- 0011
- 000111
- ...

CFG for $\{0^n 1^n | n \geq 0\}$

- A CFG for the language $\{0^n 1^n | n \geq 0\}$
 - $A \rightarrow \epsilon$
 - $A \rightarrow 0A1$

Logistics

- Classes
 - ?
- Office hours
 - ?
- Textbook
 - J. E. Hopcroft, R. Motwani, and J. D. Ullman, ***Introduction to Automata Theory, Languages, and Computation***
- Reference books
 - M. Sipser, ***Introduction to the Theory of Computation***
 - H. R. Lewis and C. H. Papadimitriou, ***Elements of the Theory of Computation***